

Graphs of functions - Set 2

TMUA practice questions by topic. Specification MM8.1, MM8.5-MM8.7, M4.15. 6 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The curves $y = 2^x$ and $y = x^2$ intersect at how many points?
- A 3
 - B 0
 - C 2
 - D 4
 - E 1
- 2 For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?
- A 1
 - B 0
 - C 2
 - D 3
 - E infinitely many
- 3 The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4 - x}$. Which sequence of transformations achieves this?
- A translate 4 units in the positive x -direction, then reflect in the y -axis
 - B translate 4 units in the negative x -direction, then reflect in the y -axis
 - C reflect in the x -axis, then translate 4 units in the positive y -direction
 - D stretch parallel to the x -axis with factor 4
 - E reflect in the y -axis, then translate 4 units in the positive x -direction

- 4 The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?
- I $a < 0$
 - II $b \neq 0$
 - III The curve has two turning points.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
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- 5 The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?
- I The equation $f(x) = g(x)$ has exactly three real solutions.
 - II The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
 - III $f(x) - g(x)$ is a polynomial of degree at least 3.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 6** $p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?
- I** $p(x) = 0$ has at least one real root.
 - II** $p(x) = 0$ has at most four real roots.
 - III** The graph of $y = p(x)$ has at least one turning point.
- A** none of them
 - B** I only
 - C** II only
 - D** III only
 - E** I and II only
 - F** I and III only
 - G** II and III only
 - H** I, II and III